

Pixelstorm-F: an open-source SIMT GPU for FPGA boards from \$30

Reference test board: Digilent Basys 3 (amazon.com/dp/B00NUE1WOG). Verilog GPU with a CUDA-style instruction set, block-RAM global memory, live VGA or HDMI framebuffer with an on-screen status line, and an on-board kernel loader. Runs on the Sipeed Tang Nano 20K (Gowin GW2AR-18, open-source tools), the Digilent Basys 3 and the Nexys A7-100T. Verified pixel for pixel against a cycle-exact golden model.

- SIMT GPU core: 1 Streaming Multiprocessor with 32-thread blocks as 4 warps x 8 lanes (Nexys A7), 8 x 4 (Basys 3) or 16 x 2 (Tang Nano 20K)
- 40-instruction PS-ISA: integer and Q16.16 arithmetic, predicates, shuffles, votes, barriers, atomics, shared memory
- Per-lane register banks (16 x 32-bit registers per thread) mapped to distributed RAM; 8 shared-memory banks
- 32 KB global memory in block RAM with a second read port for video scan-out
- Hardware loader: 4 kernels in ROM (gradient shader, triangle rasterizer, Mandelbrot, Mandelbrot zoom); one button launches
- 640 x 480 @ 60 Hz video drawing the framebuffer live, 12x magnified: VGA (Digilent boards) or HDMI/DVI with an in-house TMDS encoder (Tang Nano 20K)
- On-screen status line: kernel name and the cycle count in decimal (font ROM and hardware binary-to-decimal converter)
- Slow-motion GPU clock on the Digilent boards (1/64, 1/1024, 1/16384); cycle count on the 7-segment display and a UART status line on all boards
- Kernel upload over the USB serial port: run new kernels without a new bitstream (./pixelstorm upload)
- Hardware performance counters (instructions, lane-operations, memory transactions) with SIMD efficiency on screen and on the serial line
- Animation mode: re-launch every frame with the frame number and a live switch argument in the constant bank
- Fully open-source build for the Tang Nano 20K (Yosys, nextpnr-himbaechel, gowin_pack, openFPGALoader); Vivado scripts for the Artix-7 boards
- Same RTL as the simulated and SkyWater 130 nm versions; every kernel verified cycle for cycle against the golden model

Block diagram

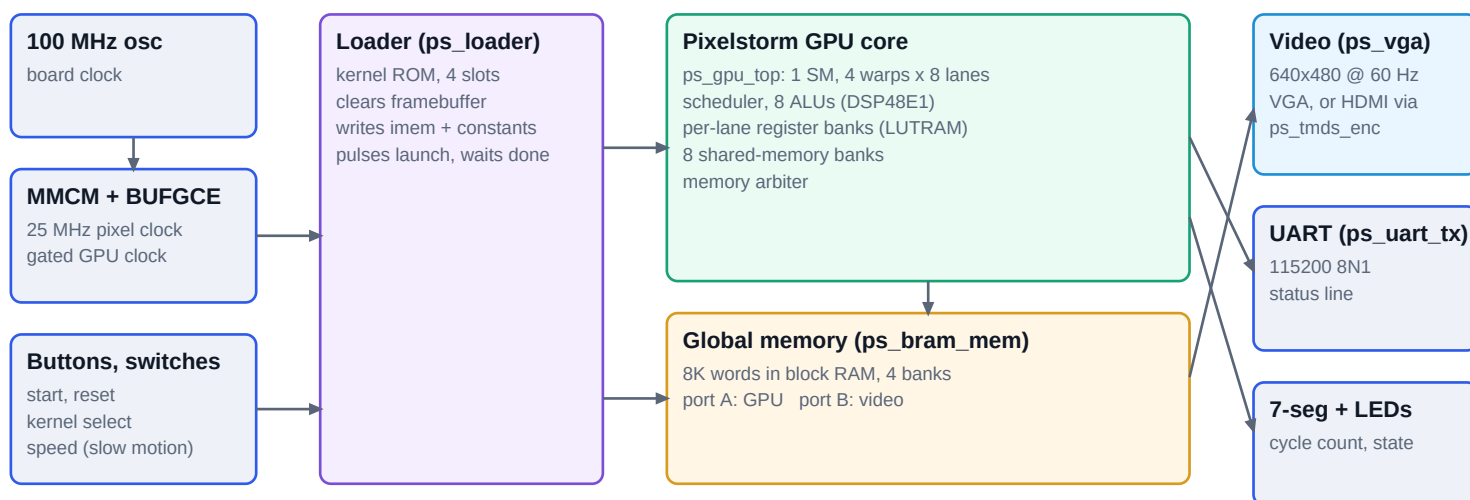


Figure 1. Pixelstorm-F top level (fpga/rtl/ps_fpga_top.v and a board wrapper).

General description

Pixelstorm-F puts the Pixelstorm GPU on an FPGA evaluation board. A small hardware loader plays the role of a host driver: it clears the framebuffer in global memory, copies the selected kernel's instructions and arguments into the GPU, and launches it. The GPU writes 0x00RRGGBB pixels to a framebuffer in block RAM while the VGA controller reads the same memory through a second port, so the image appears on the monitor as the warps compute it. The GPU core is the unmodified `ps_gpu_top` from the main repository, configured by parameters.

Configurations and resources

Parameter	Tang Nano 20K	Basys 3	Nexys A7-100T
FPGA	GW2AR-LV18QN88C8/I7	XC7A35T-1CPG236C	XC7A100T-1CSG324C
Approximate price	about \$30	about \$150 to \$200	about \$300 to \$350
Warps x lanes (1 SM)	16 x 2	8 x 4	4 x 8
Registers per thread	16 x 32 bit	16 x 32 bit	16 x 32 bit
Shared memory	256 words, 2 banks	256 words, 4 banks	256 words, 8 banks
Global memory	8K words block SRAM	8K words block RAM	8K words block RAM
Video	HDMI (DVI) 640x480	VGA 640x480	VGA 640x480
Clocks	27 MHz -> 126 MHz TMDS, 25.2 MHz GPU	100 MHz -> 25 MHz	100 MHz -> 25 MHz
Toolchain	open source or Gowin EDA	Vivado ML Standard	Vivado ML Standard

Resource estimate

Resource	Nexys A7 core	of XC7A100T	Basys 3 core	of XC7A35T
LUT	30,508	48% of 63,400	15,032	72% of 20,800
FF	1,807	1% of 126,800	1,377	3% of 41,600
DSP48E1	80	33% of 240	40	44% of 90
BRAM36	1	1% of 135	1	2% of 50

GPU core only, from yosys synth_xilinx -family xc7: Nexys A7 build = 1 SM, 4 warps x 8 lanes; Basys 3 build = 1 SM, 8 warps x 4 lanes; both 16 registers per thread. LUT includes distributed-RAM primitives (RAM64M/RAM32M, 4 LUTs each). The loader, VGA, UART and the 32 KB block-RAM global memory add about 9 RAMB36 and a few hundred LUTs. Vivado usually packs 10 to 30 percent tighter; its utilization.rpt from make fpga-bit is authoritative. Before the register file and shared memory were banked per lane, the Nexys core needed 164,482 LUTs and 26,447 flip-flops.

Tang Nano 20K: not measured on the 4 GB build machine (Yosys synth_gowin ran out of memory). make fpga-tang prints Yosys's cell counts (build/fpga/tangnano20k/utilization.txt) and nextpnr's device utilisation; check them before loading the board.

Memory map (global memory, word addresses)

Range	Contents
0x0000 - 0x0FFF	free for kernel data
0x1000 - 0x13FF	framebuffer, one 0x00RRGGBB word per pixel (32 x 32 or 32 x 16)
0x1400 - 0x1FFF	free

Kernel ROM

Slot	Kernel	Instr.	Launch	Basys 3 (8 x 4)	Nexys A7 (4 x 8)	Tang Nano (16 x 2)
0	gradient_shader	20	<<<32, 32>>>	28,000 (1.12 ms)	15,072 (0.60 ms)	55,904 (2.22 ms)
1	triangle_raster	46	<<<32, 32>>>	61,280 (2.45 ms)	31,712 (1.27 ms)	122,464 (4.86 ms)
2	mandelbrot	47	<<<16, 32>>>	81,660 (3.27 ms)	46,216 (1.85 ms)	154,252 (6.12 ms)
3	mandelbrot_zoom	47	<<<16, 32>>>	138,885 (5.56 ms)	79,461 (3.18 ms)	254,127 (10.08 ms)

Slot 3 runs the same instructions as slot 2 with different kernel arguments in the constant bank: the view zooms onto the top bulb of the set.

Pin functions

Basys 3 (reference board)

Signal	Package pin(s)	Function
clk	W5	100 MHz input
btnC	U18	start kernel
btnU	T18	reset
sw	sw[0]=V17, sw[1]=V16, sw[14]=T1, sw[15]=R2	sw1..0 kernel, sw15..14 speed
led	led[14]=P1, led[15]=L1	led15 busy, led14 done, led11 upload OK
vgaRed	vgaRed[0]=G19, vgaRed[1]=H19, vgaRed[2]=J19, vgaRed[3]=N19	VGA red
vgaBlue	vgaBlue[0]=N18, vgaBlue[1]=L18, vgaBlue[2]=K18, vgaBlue[3]=J18	VGA blue
vgaGreen	vgaGreen[0]=J17, vgaGreen[1]=H17, vgaGreen[2]=G17, vgaGreen[3]=D17	VGA green
Hsync	P19	horizontal sync
Vsync	R19	vertical sync
RsTx	A18	status line to PC
RsRx	B18	kernel upload from PC

Nexys A7-100T

Signal	Package pin(s)	Function
CLK100MHZ	E3	100 MHz input
CPU_RESETN	C12	reset, active low
BTNC	N17	start kernel
SW	SW[0]=J15, SW[1]=L16, SW[14]=U11, SW[15]=V10	SW1..0 kernel, SW15..14 speed
LED	LED[14]=V12, LED[15]=V11	LED15 busy, LED14 done
VGA_R	VGA_R[0]=A3, VGA_R[1]=B4, VGA_R[2]=C5, VGA_R[3]=A4	VGA red
VGA_G	VGA_G[0]=C6, VGA_G[1]=A5, VGA_G[2]=B6, VGA_G[3]=A6	VGA green
VGA_B	VGA_B[0]=B7, VGA_B[1]=C7, VGA_B[2]=D7, VGA_B[3]=D8	VGA blue
VGA_HS	B11	horizontal sync
VGA_VS	B12	vertical sync
UART_RXD_OUT	D4	status line to PC

Tang Nano 20K

Signal	Package pin(s)	Function
I_clk	4	27 MHz crystal
I_s1	88	S1: start kernel
I_s2	87	S2: next kernel
O_led	O_led[0]=15, O_led[1]=16, O_led[2]=17, O_led[3]=18, O_led[4]=19, O_led[5]=20	kernel, busy, done (active low)
O_uart_tx	69	status line to PC
I_uart_rx	70	
O_tmds_clk_p	33/34	HDMI TMDS clock pair
O_tmds_data_p	O_tmds_data_p[0]=35/36, O_tmds_data_p[1]=37/38, O_tmds_data_p[2]=39/40	HDMI TMDS data pairs 0, 1, 2 (blue, green, red)

All listed pins are 3.3 V LVCMOS33 (Nexys A7 SW8 and SW9 are 1.8 V and unused). Pin data comes from the constraint files in this repository, which follow Digilent's master XDC files.

Operation

Step	What to do	What you see
1	Program the bitstream (make fpga-prog or Vivado Hardware Manager)	VGA shows an empty blue frame
2	Choose a kernel with the two lowest switches	LED13..12 show the choice after start
3	Optional: slow motion with the two highest switches	10 = about 2 s for Mandelbrot
4	Press the centre button	amber bar; image paints in; LED15 on
5	Wait for done	green bar; LED14 on; 7-segment shows cycles; status rows show cycles, INSTR, SIMD and MEM; UART prints them
6	Optional: sw2 = 1 (animation), sw13..sw3 = live argument	the kernel re-runs every frame with c[15] = frame, c[14] = switches

Kernel upload

Slot 3 can be rewritten from the PC over the same serial port: `./pixelstorm upload kernel.psa --port /dev/ttyUSB1`. The board launches the kernel as soon as the checksum matches; a bad packet is ignored and a pause of about 50 ms restarts the receiver.

Bytes	Field
4	magic "PSK1"
16	kernel name (shown on the status line)
2 + 2	grid, block (little-endian)
2 + 1 + 1	framebuffer address, width, height
2	instruction count n (at most 256)
4 n	instructions
64	16 kernel arguments (constant bank)
1	checksum: sum of all bytes after the magic, mod 256

Verified in simulation on the Basys 3 shape: rings (fpga/kernels/rings.psa) uploaded at 115200 baud, 45,920 GPU cycles, PIXELSTORM k=3 cycles=0000B360, pixels PASS. PASS: corrupted upload rejected (checksum error, nothing launched)

UART status line

115200 baud, 8 data bits, no parity, 1 stop bit. After every kernel: `PIXELSTORM k=<slot> cycles=<hex> instr=<hex> lanes=<hex> mem=<hex>` followed by CR LF: the cycle count and the three performance counters, eight hex digits each. Open it with any terminal program (PuTTY, screen, minicom) on the board's USB serial port.

Video timing

Parameter	Horizontal	Vertical
Visible	640 pixels	480 lines
Front porch / sync / back porch	16 / 96 / 48	10 / 2 / 33
Total	800 pixels	525 lines
Sync polarity	negative	negative
Pixel clock	25.0 MHz (VESA 25.175 MHz, accepted by monitors)	59.5 Hz frame rate

Verification

Nexys A7 shape (4 warps x 8 lanes) first; the reference Basys 3 shape and the Tang Nano shape follow.

Before any hardware, the complete FPGA design (loader, GPU, block-RAM memory, VGA, UART) runs in Icarus Verilog (fpga/sim/tb_fpga.v). The testbench presses start, waits for done, captures one full VGA frame from the video pins, and fpga/sim/check.js compares every framebuffer pixel on screen, at its left edge, centre and right edge, with the golden model. The

GPU core itself passes the repository's cycle-exact RTL-versus-model suite.

Kernel	GPU cycles	UART line (simulated)	Pixels checked	Result
gradient_shader	15,072	PIXELSTORM k=0 cycles=00003AE0	1,024	PASS
triangle_raster	31,712	PIXELSTORM k=1 cycles=00007BE0	1,024	PASS
mandelbrot	46,216	PIXELSTORM k=2 cycles=0000B488	512	PASS
mandelbrot_zoom	79,461	PIXELSTORM k=3 cycles=00013665	512	PASS

Reference board: the same kernels on the Basys 3 shape (8 warps x 4 lanes)

Kernel	GPU cycles	UART line (simulated)	Pixels checked	Result
gradient_shader	28,000	PIXELSTORM k=0 cycles=00006D60	1,024	PASS
triangle_raster	61,280	PIXELSTORM k=1 cycles=0000EF60	1,024	PASS
mandelbrot	81,660	PIXELSTORM k=2 cycles=00013EFC	512	PASS
mandelbrot_zoom	138,885	PIXELSTORM k=3 cycles=00021E85	512	PASS

Performance counters on the Basys 3 shape (board = golden model, exactly)

Kernel	Warp instructions	Lane-operations	SIMD efficiency	Memory transactions
gradient_shader	5,120	20,480	100.0%	256
triangle_raster	11,776	46,747	99.2%	256
mandelbrot	16,092	52,382	81.4%	128
mandelbrot_zoom	27,537	84,696	76.9%	128

Animation: 3 frames with switches = 100, final frame and counters equal the golden model run with c[15] = 2, c[14] = 100: PASS.

Same kernels on the Tang Nano 20K shape (16 warps x 2 lanes)

Kernel	GPU cycles	UART line (simulated)	Pixels checked	Result
gradient_shader	55,904	PIXELSTORM k=0 cycles=0000DA60	1,024	PASS
triangle_raster	122,464	PIXELSTORM k=1 cycles=0001DE60	1,024	PASS
mandelbrot	154,252	PIXELSTORM k=2 cycles=00025A8C	512	PASS
mandelbrot_zoom	254,127	PIXELSTORM k=3 cycles=0003E0AF	512	PASS

HDMI encoder: PASS: 24000 TMDS symbols decode exactly; running disparity stays within ± 8

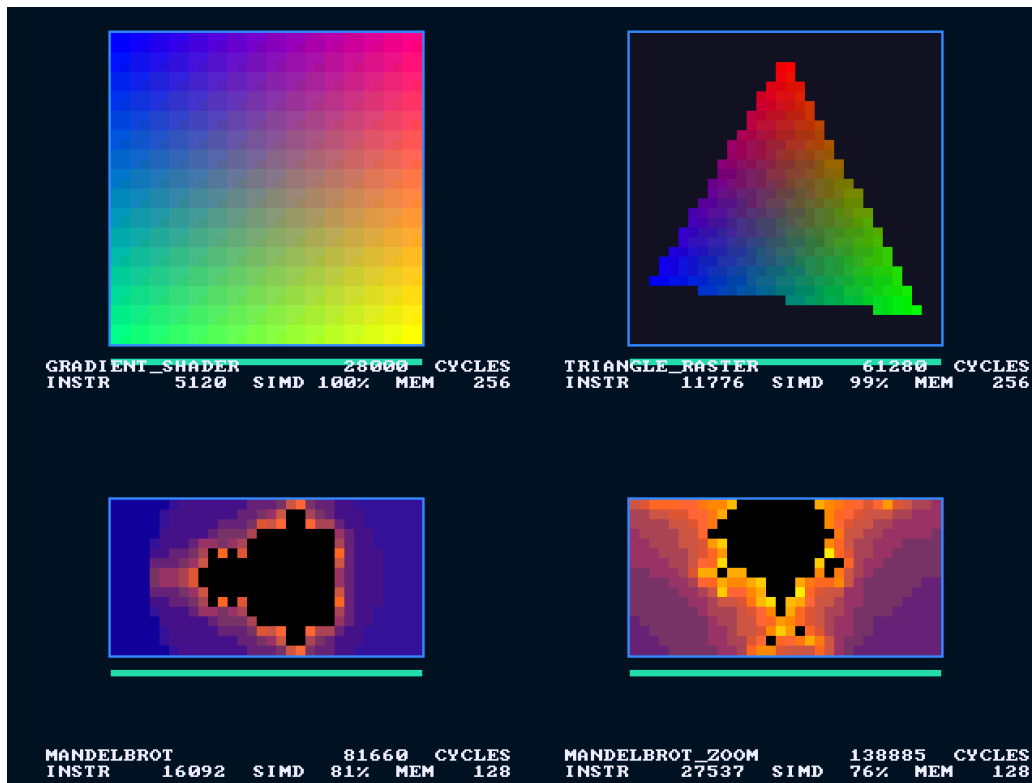


Figure 2. VGA frames captured from the simulated board: what a monitor shows for slots 0 to 3.

Build and program

Command	Result
make fpga-sim	board simulation: all four kernels on both GPU shapes with pixel checks, plus the TMDS encoder
make fpga-tang	Tang Nano 20K with the open-source flow (Yosys, nextpnr-himbaechel, gowin_pack) and load with openFPGALoader
make fpga-bit BOARD=nexys_a7	bitstream with Vivado (ML Standard, free): build/fpga/nexys_a7/pixelstorm_nexys_a7.bit, utilization and timing reports
make fpga-bit BOARD=basys3	the same for the Basys 3
make fpga-prog BOARD=nexys_a7	load over USB with openFPGALoader (or vivado -source fpga/vivado/program.tcl)

Where to buy

Board	Link	Approximate price
Sipeed Tang Nano 20K	amazon.com/dp/B0C5XJV83K	about \$30
Digilent Basys 3	amazon.com/dp/B00NUE1WOG	about \$150 to \$200
Digilent Nexys A7-100T	digilent.com (Nexys A7 page)	about \$300 to \$350

Prices change; see [fpga/docs/BUYING.md](#).

Reference documents

- Sipeed Tang Nano 20K wiki and schematic: wiki.sipeed.com/hardware/en/tang/tang-nano-20k/nano-20k.html; official examples and constraints: github.com/sipeed/TangNano-20K-example
- Gowin GW2A family data sheet and primitive user guides (rPLL, CLKDIV, OSER10, ELVDS_OBUF): gowinsemi.com
- Digilent Nexys A7 Reference Manual and schematic: digilent.com/reference/programmable-logic/nexys-a7/reference-manual
- Digilent Basys 3 Reference Manual and schematic: digilent.com/reference/programmable-logic/basys-3/reference-manual

- Digilent master constraint files: github.com/Digilent/digilent-xdc
- AMD 7 Series FPGAs Data Sheet: Overview (DS180); Artix-7 DC and AC Switching Characteristics (DS181): docs.amd.com
- AMD 7 Series user guides: CLB (UG474), Memory Resources (UG473), Clocking (UG472), DSP48E1 (UG479): docs.amd.com

Status and limits

The design is verified in simulation and its resources are estimated with Yosys; timing closure and on-board behaviour must be confirmed with Vivado's reports and a physical board. Electrical characteristics, power and ratings are those of the evaluation board: see the vendor documents above.

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